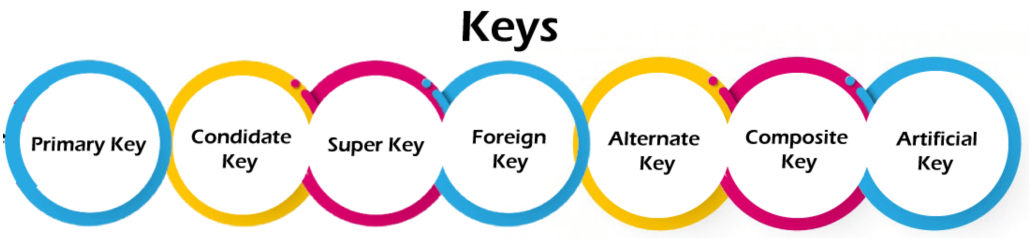
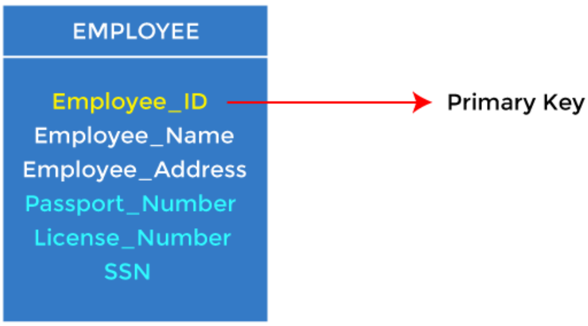
1. Explain different types of Keys.

Types of keys:



1. Primary key

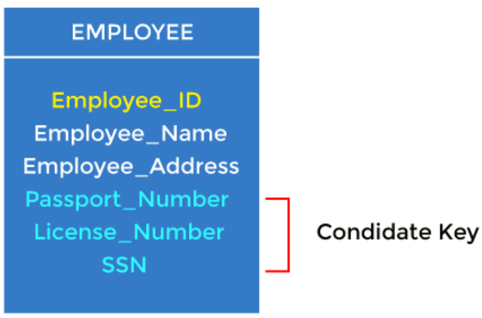
- It is the first key used to identify one and only one instance of an entity uniquely. An entity can contain multiple keys, as we saw in the PERSON table. The key which is most suitable from those lists becomes a primary key.
- In the EMPLOYEE table, ID can be the primary key since it is unique for each employee. In the EMPLOYEE table, we can even select License_Number and Passport_Number as primary keys since they are also unique.
- For each entity, the primary key selection is based on requirements and developers.



2. Candidate key

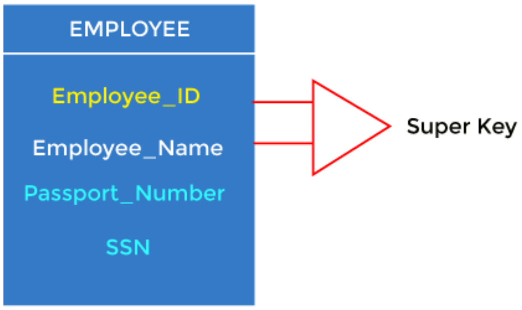
- A candidate key is an attribute or set of attributes that can uniquely identify a tuple.
- Except for the primary key, the remaining attributes are considered a candidate key. The candidate keys are as strong as the primary key.

For example: In the EMPLOYEE table, id is best suited for the primary key. The rest of the attributes, like SSN, Passport_Number, License_Number, etc., are considered a candidate key.



3. Super Key

Super key is an attribute set that can uniquely identify a tuple. A super key is a superset of a candidate key.

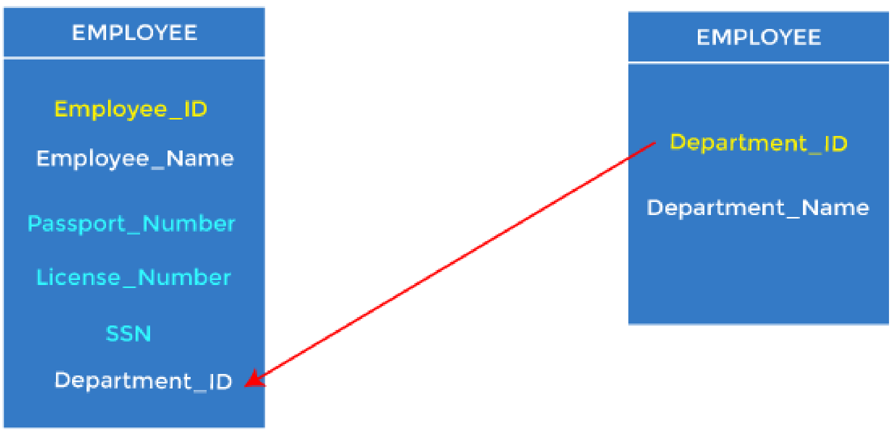


For example: In the above EMPLOYEE table, for(EMPLOEE_ID, EMPLOYEE_NAME), the name of two employees can be the same, but their EMPLYEE_ID can't be the same. Hence, this combination can also be a key.

The super key would be EMPLOYEE-ID (EMPLOYEE_ID, EMPLOYEE-NAME), etc.

4. Foreign key

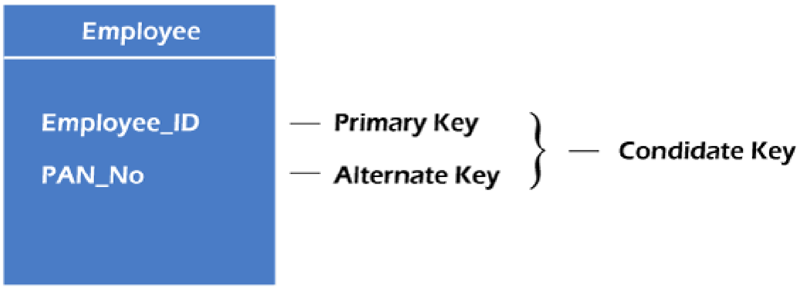
- Foreign keys are the column of the table used to point to the primary key of another table.
- Every employee works in a specific department in a company, and employee and department are two different entities. So we can't store the department's information in the employee table. That's why we link these two tables through the primary key of one table.
- We add the primary key of the DEPARTMENT table, Department_Id, as a new attribute in the EMPLOYEE table.
- In the EMPLOYEE table, Department_Id is the foreign key, and both the tables are related.



5. Alternate key

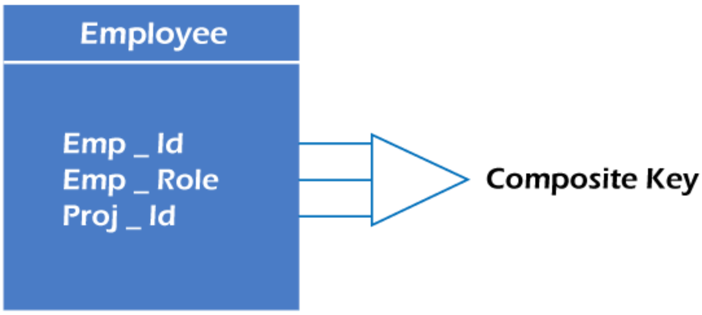
There may be one or more attributes or a combination of attributes that uniquely identify each tuple in a relation. These attributes or combinations of the attributes are called the candidate keys. One key is chosen as the primary key from these candidate keys, and the remaining candidate key, if it exists, is termed the alternate key. **In other words,** the total number of the alternate keys is the total number of candidate keys minus the primary key. The alternate key may or may not exist. If there is only one candidate key in a relation, it does not have an alternate key.

For example, employee relation has two attributes, Employee_Id and PAN_No, that act as candidate keys. In this relation, Employee_Id is chosen as the primary key, so the other candidate key, PAN_No, acts as the Alternate key.

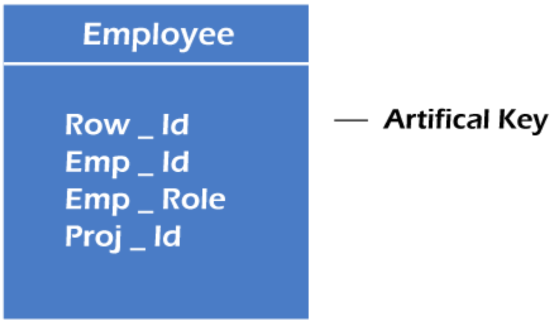


6. Composite key

Whenever a primary key consists of more than one attribute, it is known as a composite key. This key is also known as Concatenated Key.



For example, in employee relations, we assume that an employee may be assigned multiple roles, and an employee may work on multiple projects simultaneously. So the primary key will be composed of all three attributes, namely Emp_ID, Emp_role, and Proj_ID in combination. So these attributes act as a composite key since the primary key comprises more than one attribute.



7. Artificial key

The key created using arbitrarily assigned data are known as artificial keys. These keys are created when a primary key is large and complex and has no relationship with many other relations. The data values of the artificial keys are usually numbered in a serial order.

For example, the primary key, which is composed of Emp_ID, Emp_role, and Proj_ID, is large in employee relations. So it would be better to add a new virtual attribute to identify each tuple in the relation uniquely.

2. Discuss the role of Database Administrator.

A database is a storehouse for a large amount of data, which is used in organizations and firms. A **database management system** is a program that helps in updating, accessing, retrieving, representing, and manipulating the data in a database system.

Database administration involves every activity that is performed by an individual to ensure that a database is available when needed. Maintaining the integrity of a database is the primary goal of database administration.

A database administrator carries out several activities and operations to ensure the integrity, security, and availability of the data stored in the database.

Roles of a database administrator

1. **Database backup**:

   A database administrator has the responsibility to back up every data in the database, recurrently. This is necessary, so that operations can be restored in times of disaster or downtime.
2. **Database availability**:

   A database administrator has the responsibility of ensuring database accessibility to users from time to time.
3. **Database restore**:

A database administrator has the responsibility of restoring a file from a backup state, when there is a need for it.

4. **Database design**:

   A database administrator has the responsibility of designing a database that meets the demands of users. Hence, having knowledge of database design is crucial for an administrator.

5. **Data move**:

   A database administrator has the responsibility of moving a database set, say from a physical base to a cloud base, or from an existing application to a new application.

6. **Database upgrade**:

   A database administrator has the responsibility of upgrading database software files when there is a new update for them, as this protects software from security breaches.

7. **Database patch**:

   In times of new upgrades for database software, the database administrator has the responsibility of ensuring that the database system functions perfectly and works to close up any gaps in the new update.

8. **Database security**:

   Datasets are assets, and one major responsibility of database administrators is to protect the data and ensure adequate security in an organization's database.

9. **Capacity planning**:

   A database administrator has the responsibility of planning for increased capacity, in case of sudden growth in database need.

10. **Database monitoring**:

    A database administrator has the responsibility of monitoring the database and the movement of data in the database. Administrators provide access for users who require access to the database.

11. **Error log review**:

    A database administrator has the responsibility of interpreting the error messages sent by a database when there is a fault or bridge.

3. **Explain the features of DBMS**

   1. Minimum Redundancy and Duplication

   Because databases are used by so many people, the risks of data duplication are relatively high. But in a database management system, data files are shared which brings down data duplication and redundancy. Due to the fact that all information in a database management system occurs only once, the odds of duplication are quite low. In other words, the same data file is accessible to all the people using the database, and the changes made by any one of the users get reflected for the data file of all the users and therefore, redundancy and duplication are avoided.

2. Reduced amount of space and money spent on storage

All database management systems must save a large amount of data. However, proper data integration saves a lot of space in the database management system. Companies spend a lot of money to keep their data safe. They will save money on data storage and data entry if they have managed data to store.

3. Data Organization

In a Database Management system, a digital repository's information is structured in a clear hierarchical structure using records, tables, and objects. Every piece of information which we enter into our database will be structured in a catalog, making it easy to search and edit our records later.

4. Customization of the Database

Along with the default and required components (records, tables, or objects) that make up a database's structure, custom elements can be constructed to fit the demands of unique users. For example, Binary Large Objects or BLOBS can be used to store images in databases and mappings can be maintained between various tables to implement complex entities.

5. Data Retrieval

The database management system, or DBMS, accepts and stores data from users. Users can subsequently get their records from the database and save them as a file, print them, or display them on the screen. Data Retrieval becomes a big advantage of the database management systems as only authenticated users can fetch data from the database and unauthenticated users are denied access, thus improving the security of the data.

6. Usage Of Query Languages

A typical Database Management System allows users to utilize query languages for collecting, searching, sorting, altering, and other tasks that enable them to manipulate their database entries. An example of a famous query language is SQL (Structured Query Language). Anyone, even without the knowledge of any programming language, can access a Database Management System easily without hassle.

7. Multi User Access

Multiple users can access all forms of information contained in the same data store with a Multi-User Access Database Management System. A security feature additionally restricts some users from seeing and/or altering specific data types and only authenticated users can access the database.

8. Data Integrity is Maintained

Multiple users can access all information in a database, but only one user can edit the same piece of data at a time. This feature allows you to avoid database corruption and failure and ensures that the Integrity of Data is maintained.

9. Management of Metadata

Metadata is "data that provides information about other data", but not the content of the data, such as the text of a message or the image itself. The metadata library (or data dictionary) in DBMS database management software explains how the database is organized and what parts (objects, associated files, records, and so on) make up its structure.

10. Maintenance of a Large Database

Only a database management system can keep large databases of large corporations up to date. These databases necessitate a high level of security as well as backup and recovery capabilities. Database Management System includes all of these functionalities. It has the ability to keep a database with a large amount of data and information.

11. Data Durability

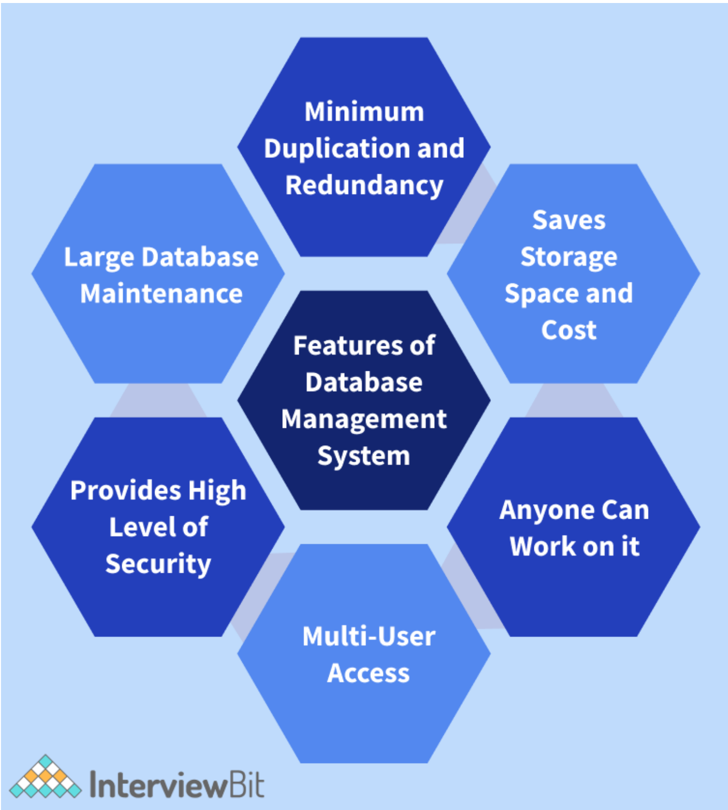
All data files are permanently stored by Database Management System, so there is no risk of data loss. If the data is lost, the organization's data files can be saved using a backup and recovery procedure. As a result, there is no need to be concerned about data loss in Database Management Systems.

12. Provides a High Level of Data Security

All companies that handle a substantial volume of data are concerned about security. Except for the Database Administrator or the department head, Database Management Systems does not grant complete database access. They have the ability to change the database and create all of the users, therefore the database management system's security level is increased.

13. Enhanced File Uniformity

Any business can build a homogeneous way to implement files and validate data uniformity with any other application programmes or systems by using the Database Management Systems. It is critical to rationalize and govern modern data management systems. A progressive database system's application software enables the application of the same rules to all data across the organization.



4. **Define data independence and explain types of data independence.**

Data Independence

- Data independence can be explained using the three-schema architecture.
- Data independence refers characteristic of being able to modify the schema at one level of the database system without altering the schema at the next higher level.

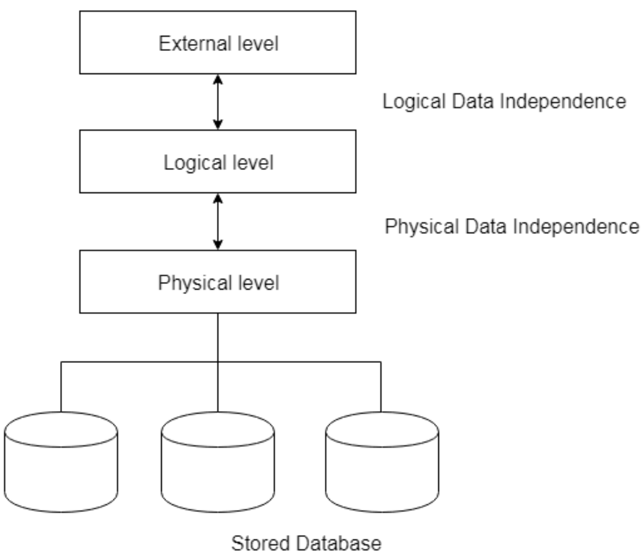
There are two types of data independence:

1. Logical Data Independence

- Logical data independence refers characteristic of being able to change the conceptual schema without having to change the external schema.
- Logical data independence is used to separate the external level from the conceptual view.
- If we do any changes in the conceptual view of the data, then the user view of the data would not be affected.
- Logical data independence occurs at the user interface level.

2. Physical Data Independence

- Physical data independence can be defined as the capacity to change the internal schema without having to change the conceptual schema.
- If we do any changes in the storage size of the database system server, then the Conceptual structure of the database will not be affected.
- Physical data independence is used to separate conceptual levels from the internal levels.
- Physical data independence occurs at the logical interface level.



5. **Explain the various advantages of DBMS over File processing system.**

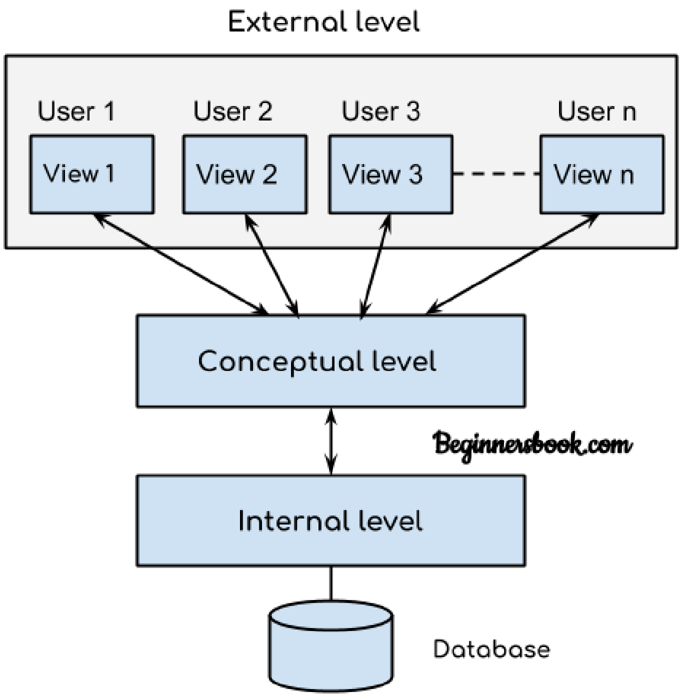
Advantages of DBMS over File system:

- **Data redundancy and inconsistency**: Redundancy is the concept of repetition of data i.e. each data may have more than a single copy. The file system cannot control the redundancy of data as each user defines and maintains the needed files for a specific application to run. There may be a possibility that two users are maintaining the data of the same file for different applications. Hence changes made by one user do not reflect in files used by second users, which leads to inconsistency of data. Whereas DBMS controls redundancy by maintaining a single repository of data that is defined once and is accessed by many users. As there is no or less redundancy, data remains consistent.
- **Data sharing**: The file system does not allow sharing of data or sharing is too complex. Whereas in DBMS, data can be shared easily due to a centralized system.
- **Data concurrency**: Concurrent access to data means more than one user is accessing the same data at the same time. Anomalies occur when changes made by one user get lost because of changes made by another user. The file system does not provide any procedure to stop anomalies. Whereas DBMS provides a locking system to stop anomalies to occur.
- **Data searching**: For every search operation performed on the file system, a different application program has to be written. While DBMS provides inbuilt searching operations. The user only has to write a small query to retrieve data from the database.
- **Data integrity**: There may be cases when some constraints need to be applied to the data before inserting it into the database. The file system does not provide any procedure to check these constraints automatically. Whereas DBMS maintains data integrity by enforcing user-defined constraints on data by itself.
- **System crashing**: In some cases, systems might have crashed due to various reasons. It is a bane in the case of file systems because once the system crashes, there will be no recovery of the data that's been lost. A DBMS will have the recovery manager which retrieves the data making it another advantage over file systems.
- **Data security**: A file system provides a password mechanism to protect the database but how long can the password be protected? No one can guarantee that. This doesn't happen in the case of DBMS. DBMS has specialized features that help provide shielding to its data.
- **Backup**: It creates a backup subsystem to restore the data if required.
- **Interfaces**: It provides different multiple user interfaces like graphical user interface and application program interface.
- **Easy Maintenance**: It is easily maintainable due to its centralized nature.

DBMS is continuously evolving from time to time. It is a powerful tool for data storage and protection. In the coming years, we will get to witness an AI-based DBMS to retrieve databases of ancient eras.

6. **Explain three-level architecture of DBMS**

DBMS Three Level Architecture Diagram



This architecture has three levels:
1. External level
2. Conceptual level
3. Internal level

1. External level

It is also called view level. The reason this level is called "view" is because several users can view their desired data from this level which is internally fetched from database with the help of conceptual and internal level mapping.

The user doesn't need to know the database schema details such as data structure, table definition etc. user is only concerned about data which is what returned back to the view level after it has been fetched from database (present at the internal level).

External level is the "top level" of the Three Level DBMS Architecture.

2. Conceptual level

It is also called logical level. The whole design of the database such as relationship among data, schema of data etc. are described in this level.

Database constraints and security are also implemented in this level of architecture. This level is maintained by DBA (database administrator).

3. Internal level

This level is also known as physical level. This level describes how the data is actually stored in the storage devices. This level is also responsible for allocating space to the data. This is the lowest level of the architecture.

7. **Explain different types of attributes in ER Model.**

Types of attributes

1. Simple attribute: It can not be divided into parts/components.

• E.g. Roll_no, phone_no, PAN_no, class etc.

2. Key attribute: It uniquely indetifies each entity in a entity set.

• E.g. Roll_no, book_id, dept_id, emp_id etc.

3. Composite attribute: It can be divided into subparts/components.

• E.g. name(fname,lname), address(latno,city,state), salary(HRA,DA,basic,TA)

4. Derived attribute: The value of derive attribute can be derived from other attributes.

• E.g. Age can be derived from DOB, percentage(marks of different subject) etc.

5. Single valued attribute: It accepts only one value for each entity instance.

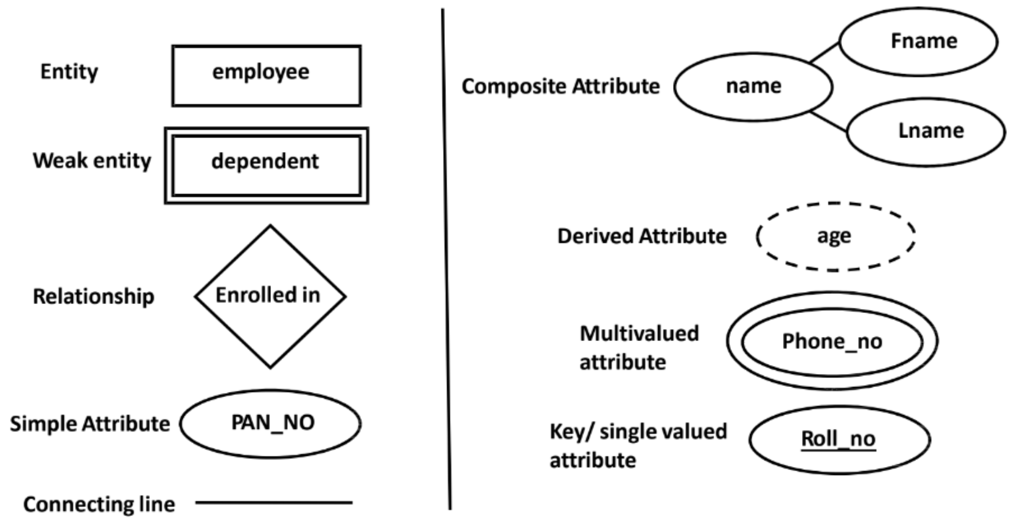
• E.g. Aadhar_no,PAN_no, roll_no, DOB etc.

6. Multivalued attribute: It accepts/has ore than one values for each entity

instance.

• E.g. Phone_no, email_id, course, degree etc.

Symbols used to draw ER diagram

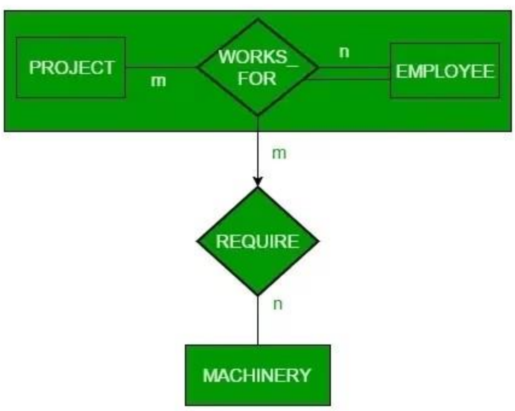


8. **Explain the following terms with an example. 1. Aggregation 2. Ternary Relationship 3. Subclass and Superclass**

Aggregation

An ER diagram is not capable of representing the relationship between an entity and a relationship which may be required in some scenarios. In those cases, a relationship with its corresponding entities is aggregated into a higher-level entity. Aggregation is an abstraction through which we can represent relationships as higher-level entity sets.

For Example, an Employee working on a project may require some machinery. So, REQUIRE relationship is needed between the relationship WORKS_FOR and entity MACHINERY. Using aggregation, WORKS_FOR relationship with its entities EMPLOYEE and PROJECT is aggregated into a single entity and relationship REQUIRE is created between the aggregated entity and MACHINERY.



Aggregation

Ternary Relationship

In Ternary relationship three different Entities takes part in a Relationship.

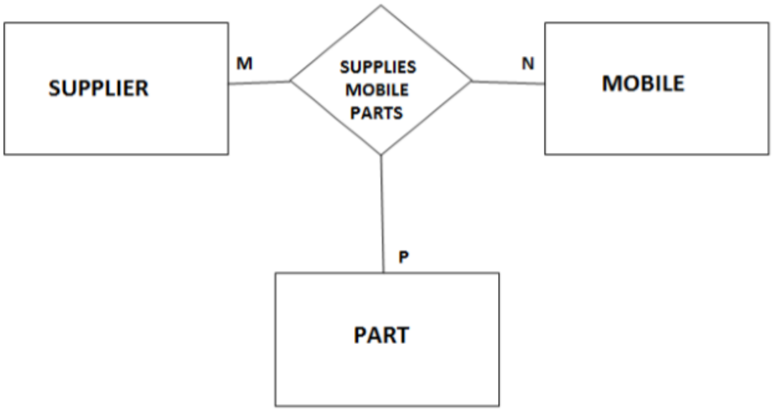
Relationship Degree = 3

For Example: Consider a Mobile manufacture company.

Three different entities involved:

- Mobile - Manufactured by company.
- Part - Mobile Part which company get from Supplier.
- Supplier - Supplier supplies Mobile parts to Company.

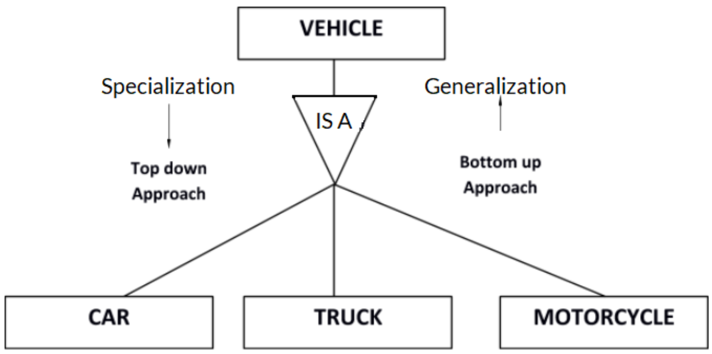
Mobile, Part and Supplier will participate simultaneously in a relationship. because of this fact when we consider cardinality we need to consider it in the context of two entities simultaneously relative to third entity.



Subclass and Superclass

Subclasses

A subclass is a class derived from the superclass. It inherits the properties of the superclass and also contains attributes of its own. An example is:



Car, Truck and Motorcycle are all subclasses of the superclass Vehicle. They all inherit common attributes from vehicle such as speed, colour etc. while they have different attributes also i.e Number of wheels in Car is 4 while in Motorcycle is 2.

Superclasses

A superclass is the class from which many subclasses can be created. The subclasses inherit the characteristics of a superclass. The superclass is also known as the parent class or base class.

In the above example, Vehicle is the Superclass and its subclasses are Car, Truck and Motorcycle.

9. Explain the Architecture of DBMS in detail.

The architecture of a Database Management System (DBMS) typically consists of three main components: the internal level, the conceptual level, and the external level.

1. **Internal Level:** This is the lowest level of the architecture and is concerned with how data is actually stored and accessed within the database. It includes the physical storage structures, indexing mechanisms, and algorithms for data retrieval and manipulation.
2. **Conceptual Level:** This is the middle level of the architecture and represents the overall structure of the database as seen by the users and applications. It includes the schema, which defines the logical structure of the database, as well as the constraints and relationships between the data.
3. **External Level:** This is the highest level of the architecture and represents the view of the database that is presented to the end users. It includes the views, which are customized representations of the data tailored to the specific needs of different users or applications.

These three levels are interconnected through a data dictionary or data catalog, which stores metadata about the database schema, constraints, and relationships. This allows for data independence, meaning that changes to the internal level do not affect the external or conceptual levels.

10. Explain the components of ER Model.

ER model stands for the Entity Relationship Model in the **database management system** (DBMS). It is the first step of designing to give the flow for a concept. It is the DFD (Data Flow Diagram) requirement of a company.

It is the basic building block for relational models. Not that much training is required to design the database project. It is very easy to convert the E-R model into a relational table or to a normalized table.

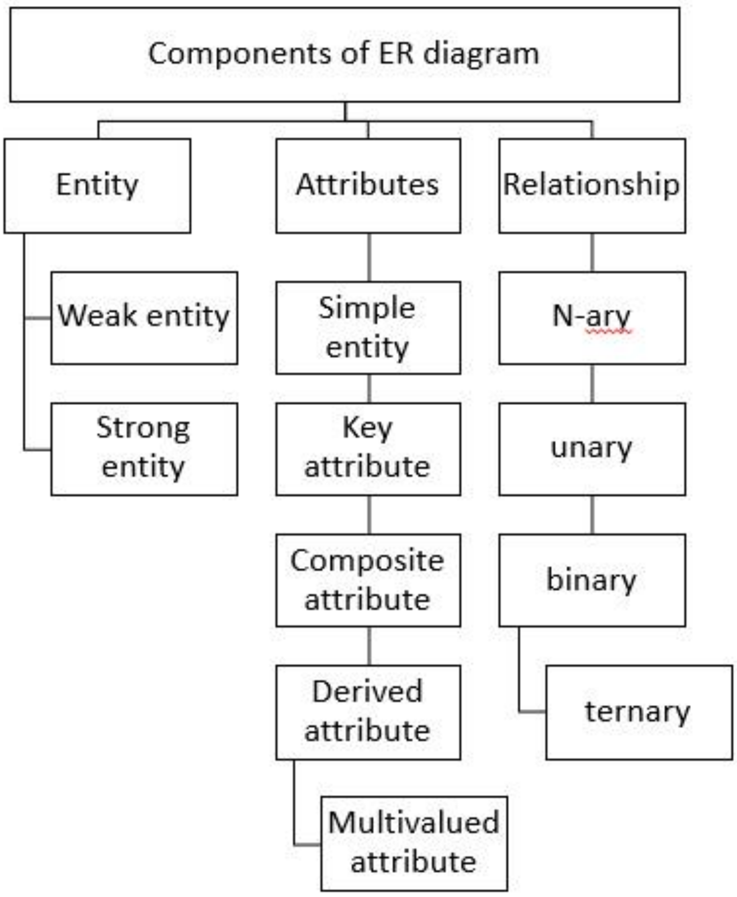
It is a high-level data model diagram that defines the conceptual view of the database. It acts as a blueprint to implement a database in future.

Components of ER diagram

The components of ER diagram are as follows −

- Entity
- Attributes
- Relationship
- Weak entity
- Strong entity
- Simple attribute
- Key attribute
- Composite attribute
- Derived attribute
- Multivalued attribute

The components of the ER diagram are pictorially represented as follows −



Entity

It may be an object, person, place or event that stores data in a database. In a relationship diagram an entity is represented in rectangle form. For example, students, employees, managers, etc.

The entity is pictorially depicted as follows −

Entity set

It is a collection of entities of the same type which share similar properties. For example, a group of students in a college and students are an entity set.

Entity is characterised into two types as follows −

- Strong entity set
- Weak entity set

Strong entity set

The entity types which consist of key attributes or if there are enough attributes for forming a primary key attribute are called a strong entity set. It is represented by a single rectangle.

For Example,

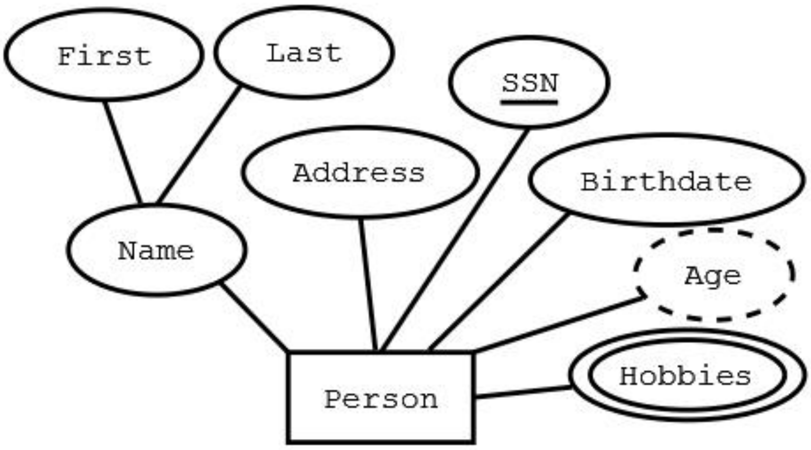
Roll no of student
EmpID of employee

Weak entity set

An entity does not have a primary key attribute and depends on another strong entity via foreign key attribute. It is represented by a double rectangle.

Attributes

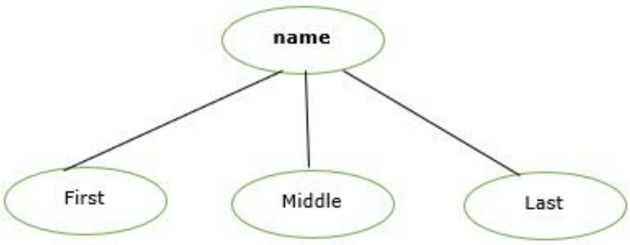
It is the name, thing etc. These are the data characteristics of entities or data elements and data fields.



Types of attributes

The types of attributes in the Entity Relationship (ER) model are as follows −

- **Single value attribute** − These attributes contain a single value. For example, age, salary etc.
- **Multivalued attribute** − They contain more than one value of a single entity. For example, phone numbers.
- **Composite attribute** − The attributes which can be further divided. For example, Name-> First name, Middle name, last name
- **Derived attribute** − The attribute that can be derived from others. For example, Date of Birth.

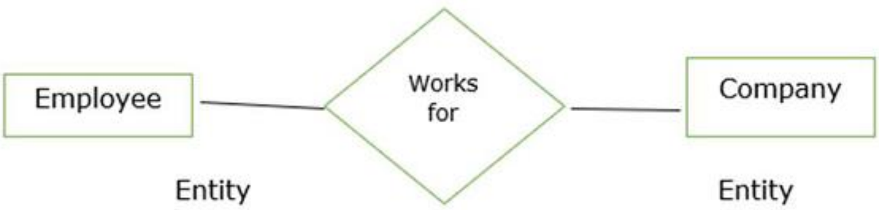


Relationship

It is used to describe the relation between two or more entities. It is represented by a diamond shape.

For Example, students study in college and employees work in a department.

The relationship is pictorially represented as follows −



Here works for is a relation between two entities.

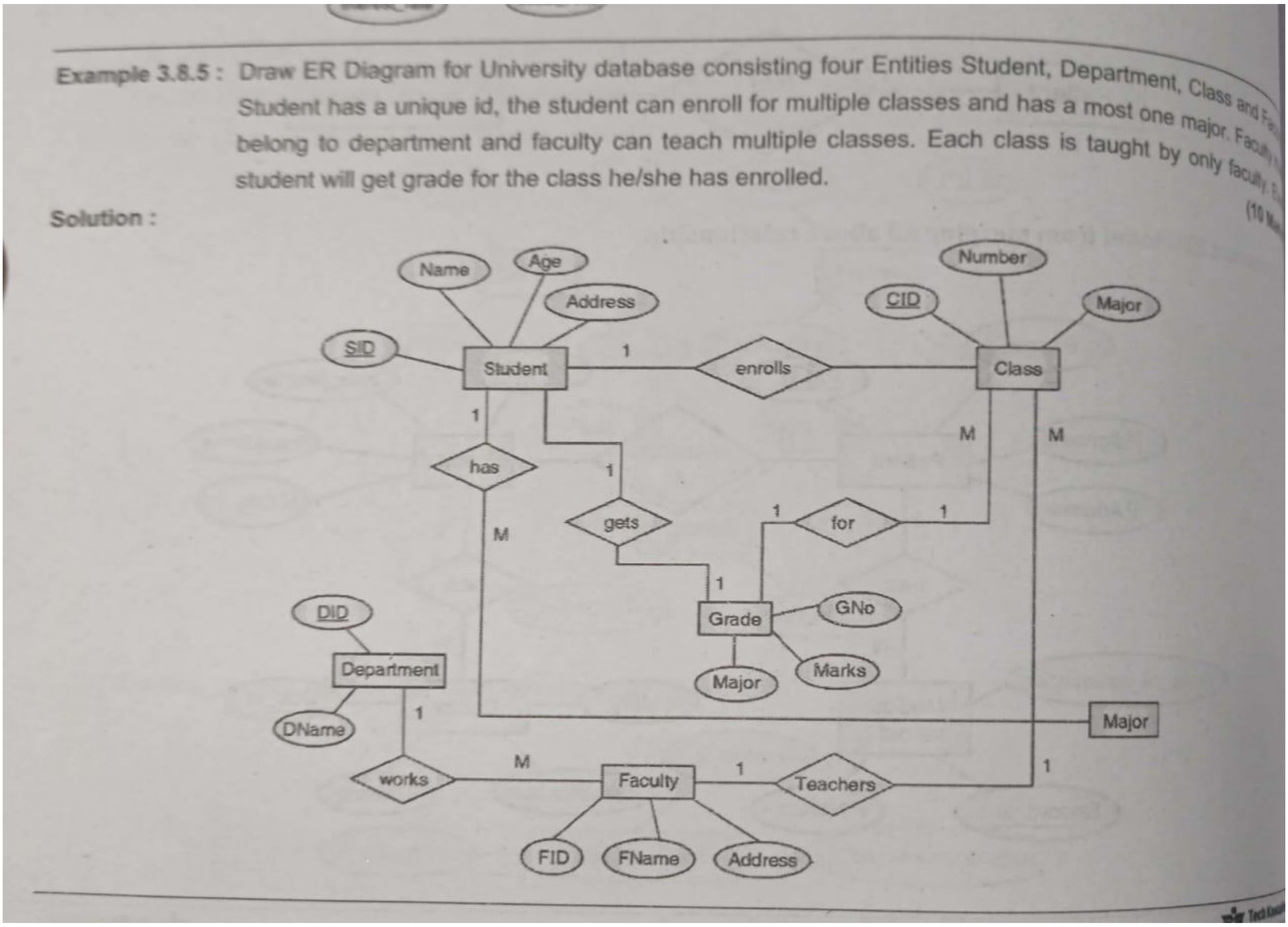
Degree of Relationship

A relationship where a number of different entities set participate is called a degree of a relationship.

It is categorised into the following −

- Unary Relationship
- Binary Relationship
- Ternary Relationship
- n-ary Relationship

11. Draw ER / EER diagram for University database.



Example 3.8.5 : Draw ER Diagram for University database consisting four Entities Student, Department, Class and
Student has a unique id, the student can enroll for multiple classes and has a most one major.
belong to department and faculty can teach multiple classes. Each class is taught by only
student will get grade for the class he/she has enrolled.

Solution :